

Sailing Boat Price Analysis with BP Neural Network Regression Summary

We have developed a **model for pricing** used sailing boats, taking into account the varying impact of different boat characteristics on pricing, as well as the important role of brokers and pricing strategies in the boat-selling process.

For the first problem, we constructed a **BP neural network regression** model to analyze the impact of various factors on pricing. Prior to this, we preprocessed the collected data using **factor analysis** to identify the key factors. Next, we applied neural network regression to **assess the weight** of each factor affecting pricing between single-hull and double-hull vessels. The results show that Catamarans are priced more on the **material** of the vessel and Monohulled Sailboats are priced more on the **year** of the vessel. Besides, The accuracy of our model is **up to 90%**.

For the second problem, we improved our model for the second problem by **introducing dummy variables** to identify the impact of **different regions** on pricing while avoiding **collinearity issues**. Substitution of data from different regions into the model, we found that the same factor is not necessarily influenced equally by different areas. Lastly we used **box plots** to display the average prices of sailboats in these three regions.

For the third problem. Firstly, we extracted a subset of informative data from the provided dataset, followed by replacing prices in the Hong Kong market. After obtaining the data, we input the processed data into our model, the result that the Material factor has a **strong influence** on the pricing of both Monohulled Sailboats and Catamarans in Hong Kong. The accuracy in practical application reached 95%.

Sensitivity analysis was conducted on the models for the first and second questions, and feature importance of each parameter was obtained. The sensitivity of the year variable was found to be **higher**, which aligns with people's tendency to prefer newer products over older ones. After introducing dummy variables, the stability of the model was **greatly improved**, indicating **strong adaptability**.

Finally, We analysis of the model results leads us to some **interesting conclusions**. In analyzing the data box plots for the Caribbean, the United States and Europe, we found that the level of **economic development** in the region was **positively** correlated with the price of catamarans and **negatively** correlated with the average price of single hulls.

Keywords: BP neural network; Regression factor analysis; Adaptive moment estimation; Pricing model

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1 Introduction

1.1 Problem Background

Sailing boats, which use wind power to move across water, are an ancient mode of transportation that came after boats and rafts. Like many luxury items, the value of a sailing boat can fluctuate with age and market conditions, and there are many factors that can affect its pricing, such as the material of the hull, the length of the boat, and the area of the sail. These influencing factors are interconnected. Understanding the composition of boat prices and their influencing factors can help operators more comprehensively assess market trends and objectively measure the true value of sailing boats. Analyzing and predicting the trend of sailing boat prices and understanding the patterns of market fluctuations in specific regions can provide a reliable basis for business decisions. Therefore, determining the pricing of a sailing boat is an important consideration.



Figure 1(Source:<http://yachter123.com/sy>)^[1]

1.2 Restatement of the Problem

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems:

- Problem 1 Considering any predictive factors you deem useful, refer to data from other sources, develop a mathematical model, and explain the listing price of each sailboat provided in the spreadsheet.
- Problem 2 using our model to explain the impact of regions on listing prices. If there is an impact, please discuss whether the regional impact is consistent across all sailboat variants, and explain the practical and statistical significance of any regional impacts you notice.
- Problem3 selecting an information-rich subset of sailboats from the provided spreadsheet, divided into monohull and catamaran sailboats. Find comparable listing

price data for this subset in the Hong Kong SAR market. Simulate the regional impact (if any) of Hong Kong SAR on the listing prices of each sailboat in your subset. Are the effects the same for catamarans and monohulls?

- Problem 4 What other interesting and meaningful conclusions or inferences did our obtain from the data?
- Problem 5 Prepare a one to two page report for sailboat brokers in Hong Kong (SAR), including several carefully selected graphs to help brokers understand your conclusions.

1.3 Literature Review

As the world's freight traffic increases, the second-hand boat market continues to develop and research on the pricing of second-hand boats continues to grow.

In "Auction Prices and Determinants of Prices for Used Sailing Yachts" (2012), AndreasSchneck and Ralf Wagner analyse auction data from the German second-hand sailboat market and find that different factors have an impact on the price of sailing yachts. In the analysis of auction data from the German used sailing yachts market, it was found that different factors influence the price of sailing yachts, with frequency of use and maintenance condition having the most significant effect on price. Xiao Qijun and Zhang Yanmeng in "A study on used ship price evaluation based on BP neural networks (2013)"^[3] developed a BP neural network model to estimate used ship prices based on age, deadweight tonnage (DWT), newbuilding price and one-year charter rate. Roberto Zanola in "Estimating the Determinants of Second-Hand Boat Prices Using Quantile Regression" (2014) use a quantile regression model to analyse data from the Italian second-hand sailboat market. In "Application of BP Neural Networks in Second-Hand Boat Evaluation (2015)"^[2],Zhang Shufan and Zhang Xin introduced BP neural networks in the evaluation process, using actual historical transaction data on clarks o n and training models to achieve the simulation of the evaluated boat prices. Feng Shan and Wu Meng in "Research on the price evaluation of used sailing boats based on fuzzy integrated evaluation method" (2016) used fuzzy integrated evaluation method to analyze the data of Chinese used sailing boat market and established an evaluation model. Liu Xia and Zhang Jie used regression analysis to analyse data from the Chinese second-hand sailing boat market and predict the future price trends of sailing boats in "A study on used sailing boat market price forecasting based on regression analysis" (2017).

As mentioned above, the pricing of used sailing boats is influenced by a number of factors, including the make, model, size, year, frequency of use and maintenance of the boat. In addition, prices vary considerably from region to region, and pricing methods vary as well. We used factor analysis to determine the main factors to ensure the accuracy of the study, and combined with neural network algorithms to build a model to obtain more informative results.

1.4 Our Work

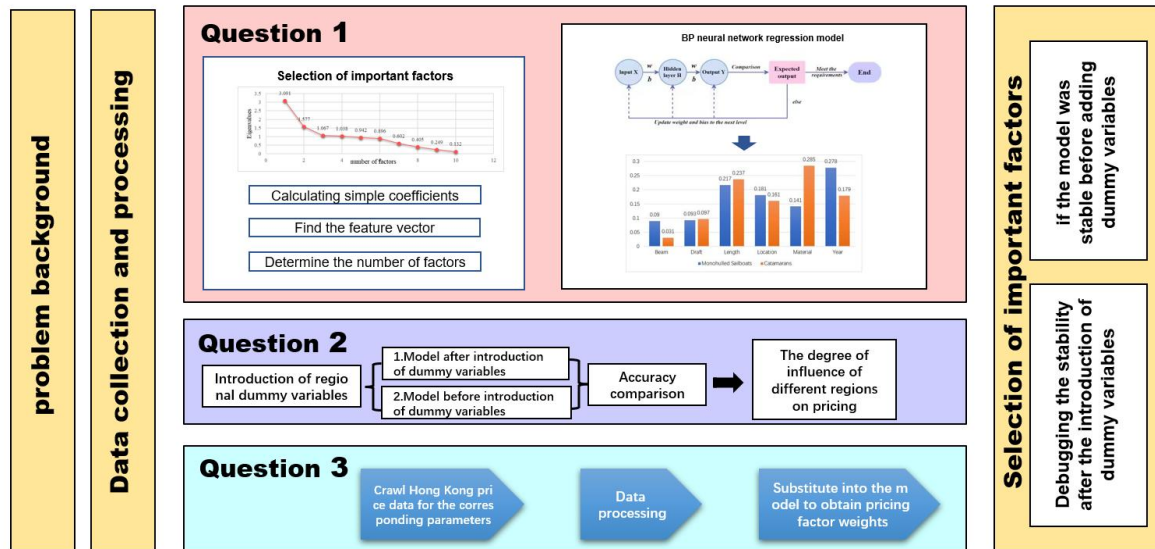


Figure 2 The Framework of Our Work

2 Assumptions and Justifications

- (1) **Assumption:** We assume the samples of parameters selected for the sailboat are independently and homogeneously distributed, i.e. the distribution of each sample is the same and independent of each other.

Justification: Sailboat pricing is determined by a combination of different factors, which are not directly dependent on each other and which may have different pricing in different markets and situations

- (2) **Assumption:** We assume that there is no perfect covariance between the parametric variables of the sailboat, i.e. that there is no identical information between the variables.

Justification: When perfect covariance exists, it means that some of these variables can be accurately represented or predicted by other variables, which can lead to multiple or unstable solutions in the model, thus making the results unreliable or uninterpretable. Therefore, to ensure the accuracy and interpretability of the model, we need to ensure that there is no identical information between the variables

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description
θ	Activation function
N_i	Number of neurons
N_o	Output layer's number of neurons
N_s	Training set's number of samples

L	Learning rate
w_i	Connection weight of the i – th Neuron
x_i	Input from the i – th Neuron

4 Data processing

4.1 Data Collection

In order to construct a model, it is necessary to gather data as the problem at hand lacks comprehensive information. To fully analyze the issue, we need to collect detailed data pertaining to the characteristics of a sailing boat. Fortunately, numerous websites dedicated to sailboat pricing offer us access to a wealth of information. For further insight, please refer to Table 2, which outlines our detailed data sources.

Table 2 Data source website

Database Names	Database Websites
Type,Engine_count	www.SailboatData.com
Price,Location,Material,Cabins,Motor_type	www.sailboatlistings.com
Length,Beam,Draft,Year	www.yachtworld.com

4.2 Data cleaning and processing

4.2.1 Removing irrelevant data

First of all, since there are missing values in the collecting class variables, e.g. Country/Region/State, random completions will affect the overall characteristics of the data, so these variables are excluded.

4.2.2 Quantitative processing

In order to facilitate further model analysis, this article has quantified three indicators: location, material, and motor type, and the quantitative results are shown in Table 3.

Table 3 Data encoding

Label	Before quantification	After quantification
	New	
Location	York,Florida,Rhode,Island,Pennsylvania.....North Dakota,West Virginia	1,2,3,...,49,50
Material	Fiberglass,wood,composite,steel,other,aluminum	1,2,3,4,5,6
Motor_Type	Motorless,inboard,outboard	1,2,3

4.2.3 Handling outliers.

To improve the accuracy and reliability of the data, it is important to detect and handle outliers using the Median Absolute Deviation (MAD) method. In the case of sailing data, outliers can be caused by measurement errors, equipment malfunctions, or other factors, and

must be addressed to ensure the integrity of the data. Therefore, employing MAD detection and processing techniques is crucial.

$$MAD = median(|x_i - x_m|) \quad (1)$$

x_i is the value in the data sample set, x_m is the median of that sample set of data.

4.2.4 Min-max normalization

Considering the subsequent establishment of the intelligent model and factor analysis model, the continuous data such as Year, Length and Beam data, are normalized by min-max to reduce the inconsistency of the dimension. What's more, This accelerates the convergence speed of intelligent algorithms and improves the stability and robustness of the model.

$$x'_i = \frac{x_i - \min(x)}{\max(x) - \min(x)} \quad (2)$$

x_i is the i th sample eigenvalue in the original dataset, x'_i is the corresponding normalized value, $\max(x)$ and $\min(x)$ are the corresponding column maximum and minimum values.

5 Pricing model based on BP neural network regression

5.1 Filtering variables by factor analysis

First, KMO and Bartlett's tests (Table 4) should be conducted to determine if factor analysis is appropriate.

Table 4 KMO test and Bartlett's test

KMO values		0.786
Approximate cardinality		18497.713
Bartlett's test of sphericity	df	45
	P	0.000***

Note: ***, **, * represent 1%, 5%, 10% level of significance respectively

KMO values between 0.7 and 0.9 indicate that this paper is suitable for factor analysis.

Factor analysis followed:

Step1: Calculate the simple coefficient matrix R between the variables

$$R_{10 \times 10} = \begin{bmatrix} r_{11} & r_{12} & \dots & r_{110} \\ r_{21} & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ r_{101} & \dots & \dots & r_{1010} \end{bmatrix} \quad (3)$$

Step2: Find the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_{10}$ of R and the corresponding eigenvectors $\mu_1, \mu_2, \dots, \mu_{10}$

Step3: Calculation of factor loadings, determining the number of factors.

$$\left\{ \begin{aligned} A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{110} \\ a_{21} & a_{22} & \dots & a_{210} \\ \dots & \dots & \dots & \dots \\ a_{101} & a_{102} & \dots & a_{1010} \end{pmatrix} &= \begin{pmatrix} u_{11}\sqrt{\lambda_1} & u_{12}\sqrt{\lambda_2} & \dots & u_{110}\sqrt{\lambda_{10}} \\ u_{21}\sqrt{\lambda_1} & u_{22}\sqrt{\lambda_2} & \dots & u_{210}\sqrt{\lambda_{10}} \\ \dots & \dots & \dots & \dots \\ u_{101}\sqrt{\lambda_1} & u_{102}\sqrt{\lambda_2} & \dots & u_{1010}\sqrt{\lambda_{10}} \end{pmatrix} \\ a_k = \sum_{i=1}^k S_i^2 / p &= \sum_{i=1}^k \lambda_i / \sum_{i=1}^{10} \lambda_i \end{aligned} \right. \quad (4)$$

We can conclude that $k=6$ by analyzing the results in Table 5, where a_k represents the cumulative contribution of feature k to the explained variance by d and R's eigenvalues should be greater than 1.

Table 5 Total variance explained

Factor	Characteristic roots	Explanation of variance	Total explanation of variance
Year	3.091	30.913	30.913
Length	1.577	15.774	46.687
Draft	1.067	10.671	57.358
Beam	1.038	10.379	67.737
Location	1.017	9.424	77.161
Material	1.001	8.961	91.122
Motor_Type	0.602	6.02	94.142

From Table 2, the cumulative contribution of the first 6 factors is $91.122\% > 90\%$. which can reflect the overall information more comprehensively.

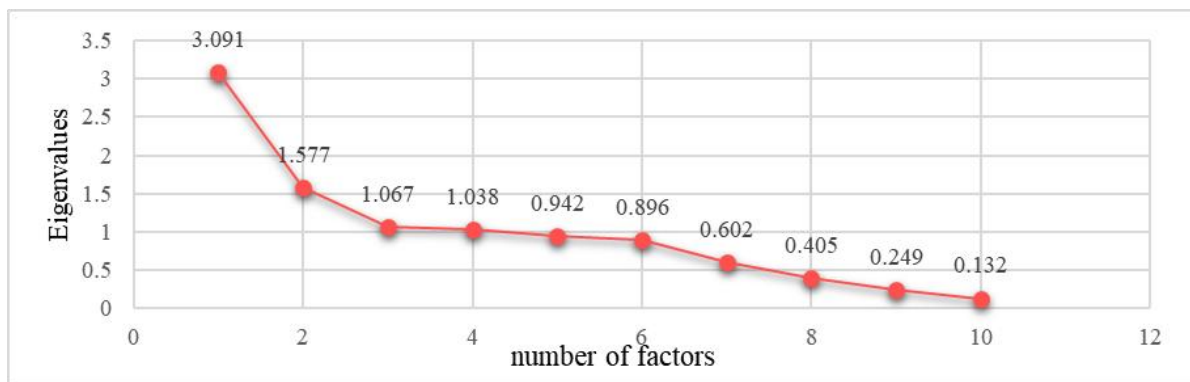


Figure 3 Rubble map

while analysis of the gravel map shows that the slope is gentler by the 6th component, so the first 6 components can be selected as the principal components. Indicates that Year,

Length, Draft, Beam, Location, Material were selected as the main factors.



Figure 4 Factor load matrix heat map

Analysis of the factor loading matrix heat map shows that the Length, Draft and Beam factors have large loading coefficients and can be identified as a particular component.

5.2 BP neural network regression model

After determining the important parameters, this study opted for a BP neural network regression model due to the complex relationship between the pricing of different types of sailboats and their respective factors. This model has strong non-linear fitting capabilities and is suitable for various types of regression analyses. Additionally, the model automatically adjusts its parameters based on the data characteristics to achieve a better fit.

The flowchart for building a model is shown below::

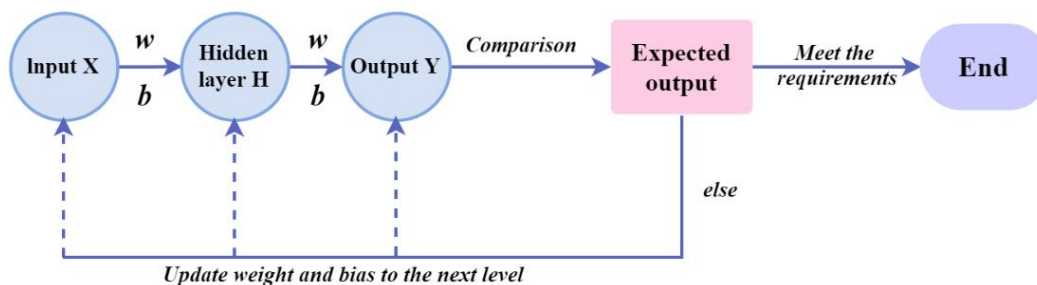


Figure 5 Model flowchart

5.2.1 neuron model:

The model takes length, draft, year, beam, location, and material as independent variables and assigns a corresponding weight to each. The values of these variables are multiplied by their respective weights and processed through an activation function to obtain the final output y , which represents the price.

Output formula:

$$y = f\left(\sum_{i=1}^n w_i x_i - \theta\right) \quad (5)$$

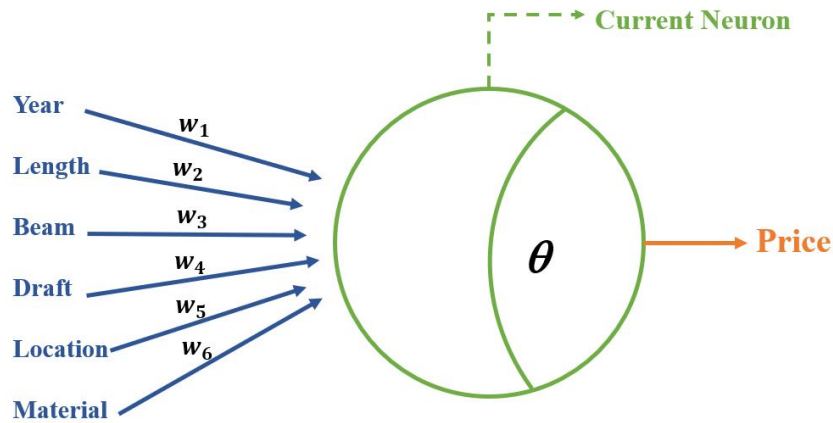


Figure 6 neuron

5.2.2 activation function:

We determined the remaining parameters using the method of controlling variables, and obtained the following results by utilizing various activation functions. we ultimately decided to use the sigmoid function.

- Sigmoid

$$\theta = \frac{1}{1 + e^{-z}} \quad (6)$$

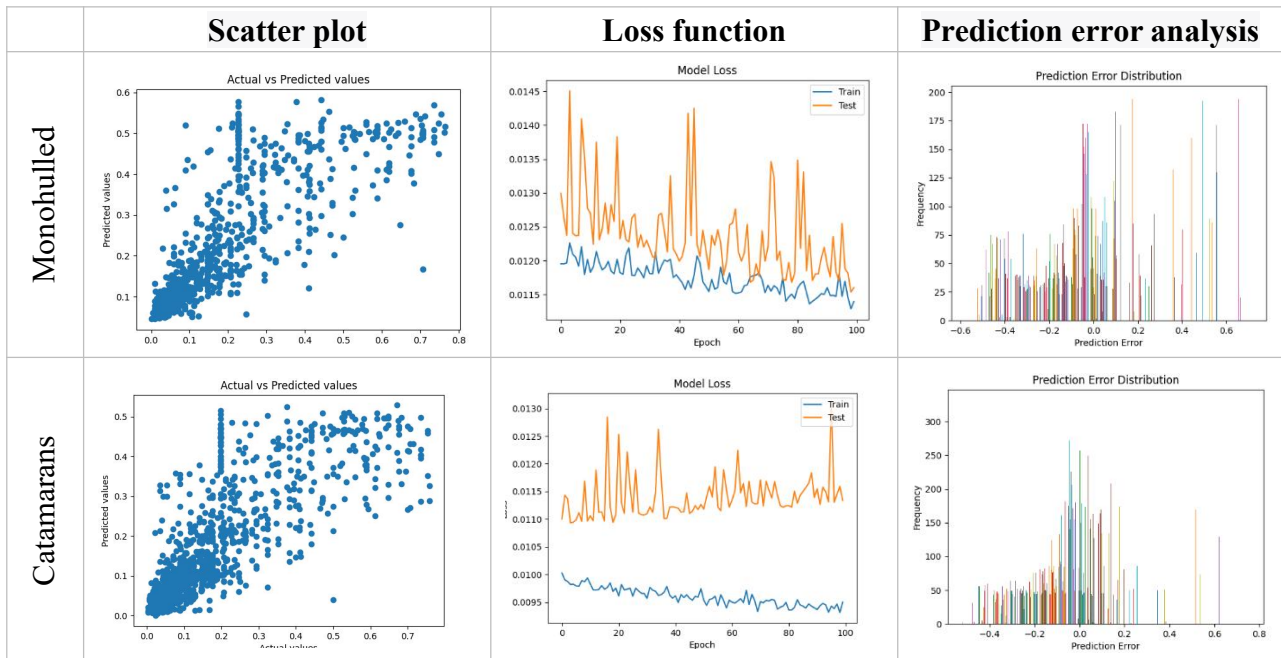


Figure 7 Comparison chart of sigmoid

- Tanh

$$\theta = \frac{e^z - e^{-z}}{e^z + e^{-z}} \quad (7)$$

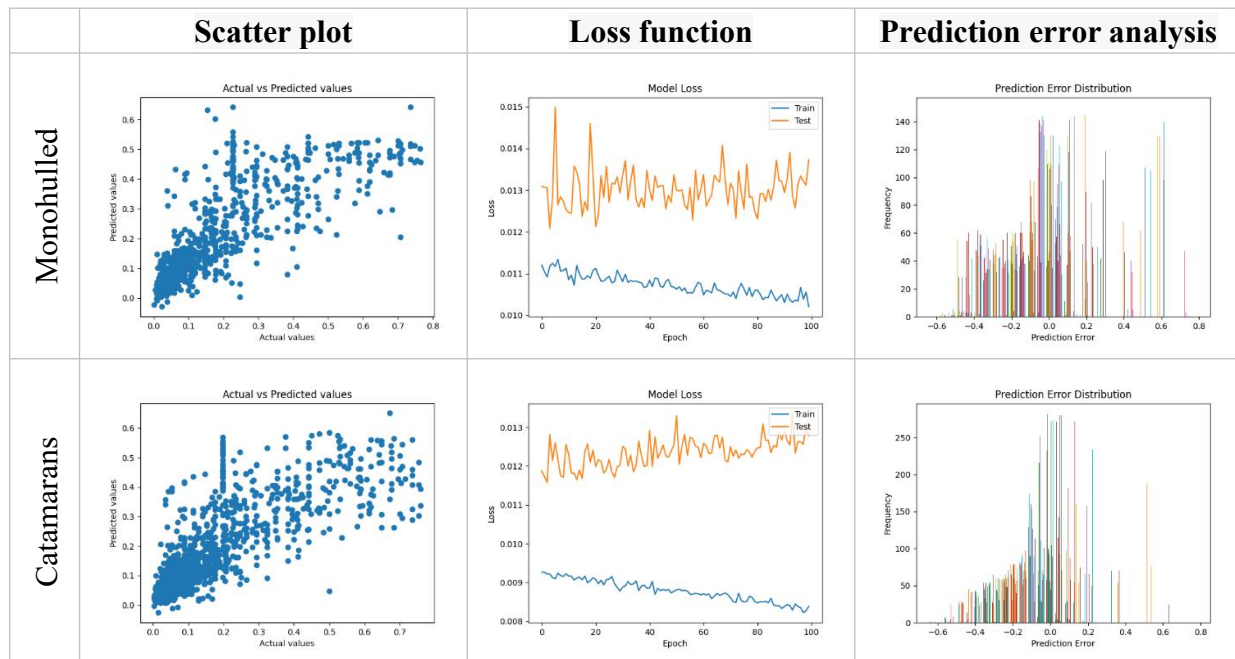


Figure 8 Comparison chart of Tanh

- Relu

$$\theta = \max(0, x) \quad (8)$$

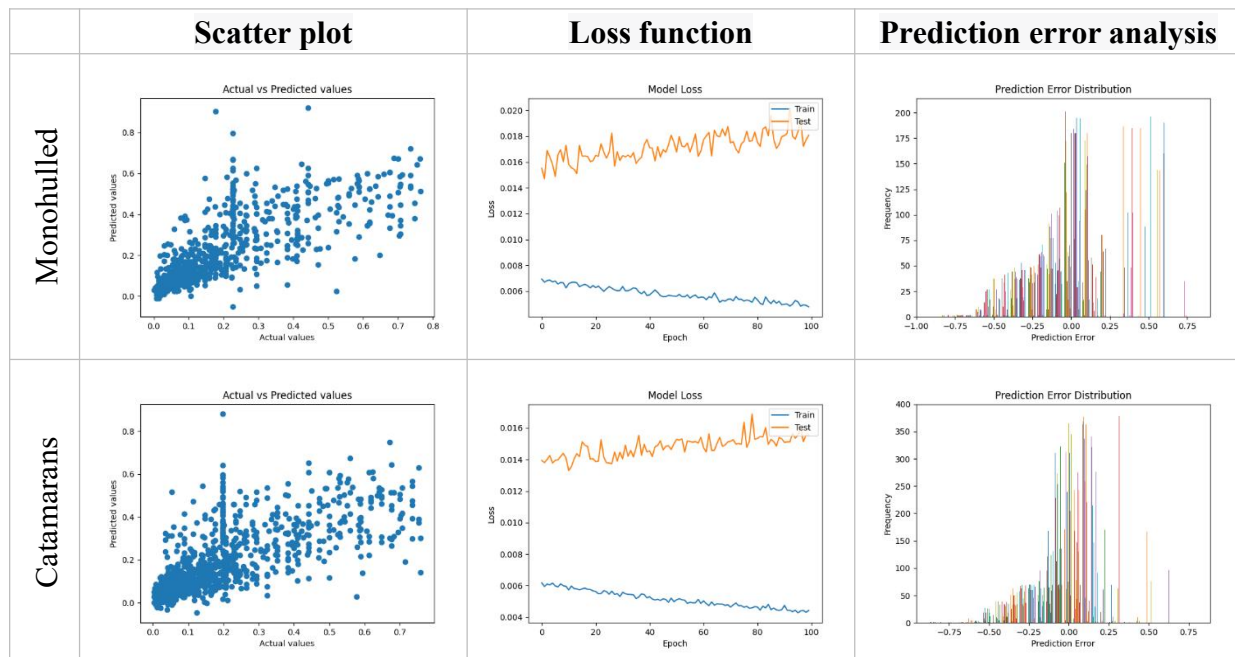


Figure 9 Comparison chart of Relu

It is evident that for both "monohull" and "catamarans" boats, the optimal solution is obtained by iterating through the sigmoid function.

5.2.3 basic framework:

This model consists of three layers.

(1) Input layer:

This layer serves as the input terminal for information, including variables such as length, draft, year, brand, location, and material.

(2) Hidden layer:

This layer is responsible for processing the information and contains a total of "q" neurons. The formula for calculating the value of "q" is as follows:

$$N_b = \frac{N_s}{(\alpha * (N_i + N_o))} \quad (9)$$

α : Arbitrary parameter, the range of 'l' value is 2-10

In this model, we take the value of q as 50.

From the above comparison, the activation function used this time is still sigmoid function.

(3) output layer:

This layer is used to output , which is the Listing Price we are looking for, Since the output layer does not need to change state, we set the activation function to itself. The function is as follow:

$$W = x' \quad (10)$$

The first step is the forward propagation of information, followed by the second step, which is the backward propagation of information. During each propagation, the weights and biases from the hidden layer to the output layer, as well as those from the input layer to the hidden layer, are adjusted in sequence. The respective input formulas are as follows:

The input of the jth input neuron:

$$W_j = \sum_{h=1}^q w_{hj} b_h \quad (11)$$

The input of the hth hidden layer neuron:

$$W_h = \sum_{i=1}^d v_{ih} x_i \quad (12)$$

5.2.4 Forward propagation:

This is the process of entering information from the input layer into the network, and obtaining the final input layer result through the calculation of each layer. We multiply the value with the weight and bring it into the activation function, the formula is as follows:

From input layer to hidden layer:

$$W_h = \sum_{i=1}^d v_{ih} x_i + \theta_h \quad (13)$$

From hidden layer to output layer:

$$W_j = \sum_{h=1}^q w_{hj} b_h + \theta_j \quad (14)$$

5.2.5 Back propagation:

Adjust the network parameters by calculating the error between the output layer and the expected value, so that we can become the error smaller. When the loss function is an error value, the formula for calculating the error is as follows:

$$E = \frac{1}{2} \sum_{k=1}^2 (y_k - T_k)^2 \quad (15)$$

For the weight update of backpropagation, in this model we use the adam algorithm to optimize the weight. This algorithm combines Momentum and Adaptive Learning Rate, Combines the average of past gradients with the learning rate in each direction. Finally, we get the updated weights:

$$W_t = W_{t-1} - \eta \nabla L(W_{t-1}) \quad (16)$$

5.2.6 consequent:

Due to the numerous weight values between neural networks, we need to list the sum of weight vectors. The specific weight values can be found in Figure 10.

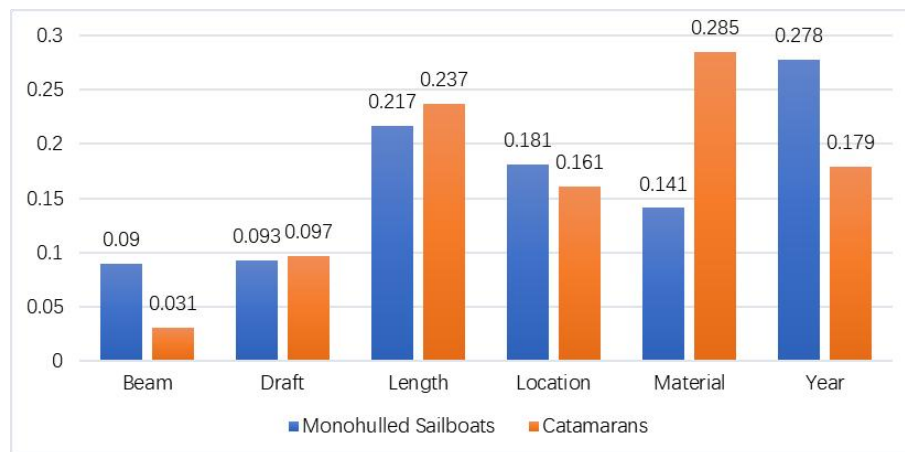


Figure 10 Weight of important factor.

As can be seen in Figure 10 Catamarans are priced more on the material of the vessel and Monohulled Sailboats are priced more on the year of the vessel.

6 With the introduction of regional dummy variables bp neural network regression

6.1 Introduction of dummy variables

Since the location data is categorical data with explicit categories represented by coded numbers, and the magnitude of the numbers does not have a comparative meaning in itself, when the location data is included in the BP neural network regression analysis, it can be treated as a dummy variable in regression analysis, which can correctly analyze the area effect and improve the accuracy of the model. As it is necessary to consider the impact of different regions on the listed area, this article divides all regions into USA, Europe, and

Caribbean, and removes the "location" variable. The transformation process is shown below:

Table 6 Dummy variable coding

Location	Is_Caribbean	Is_Europe	Is_USA
Caribbean	1	0	0
Europe	0	1	0
USA	0	0	1

6.2 bp neural network model with the introduction of dummy variables

After introducing the dummy variables, the independent variables became Length, Draft, Year, Beam, Is_Caribbean, Is_Europe, Is_USA and Material. The calculation formula for a single neuron is:

$$y = f\left(\sum_{i=1}^8 w_i x_i - \theta\right) \quad (17)$$

Single Neuron Model is shown in the following Figure 11.:

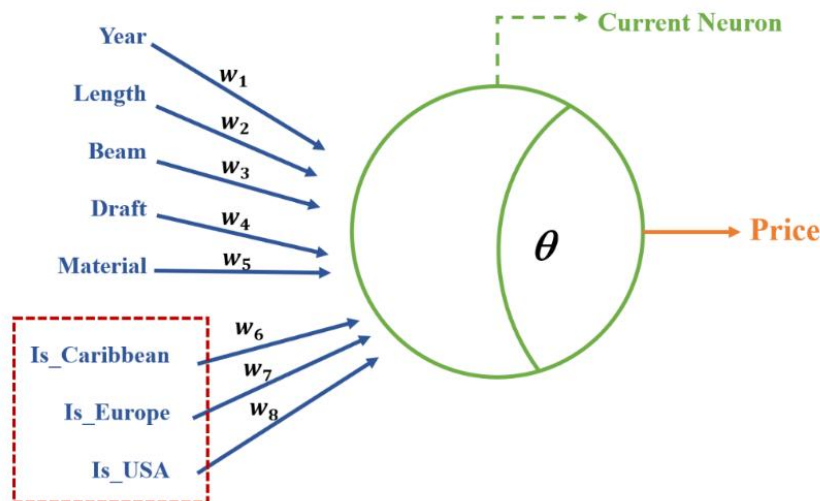


Figure 11 Neurons after adding dummy variables

We use the sigmoid function as the activation function, which has been compared in the previous text and has good accuracy. The formula is:

$$\theta = \frac{1}{1 + e^{-z}} \quad (18)$$

6.2.1 Establishing models for each layer.

For the input layer, the variables Length, Draft, Year, Beam, Is_Caribbean, Is_Europe, Is_USA, and Material are represented as X1, X2, X3, X4, X5, X6, X7, and X8 respectively, and are fed into the input layer. For the hidden layer, 64 neurons are established using the formula, with three hidden layers, as shown in the following Figure 12:

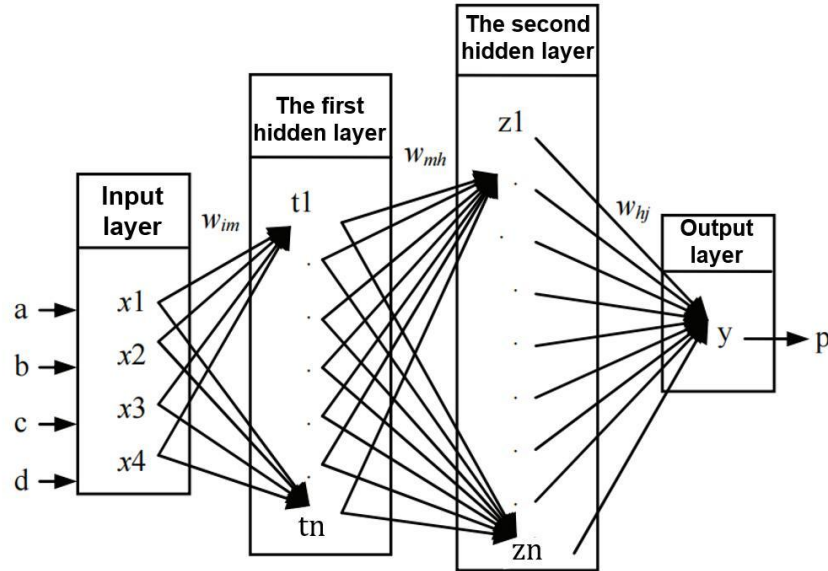


Figure 12 Principle diagram

Finally, the results obtained through iteration are presented through the output layer.

6.2.2 Determination of weights

Firstly, perform forward propagation by introducing the values of the input layer to the first hidden layer, then to the second hidden layer, and finally to the third hidden layer. Finally, output to the output layer. The formula is as follows:

From input layer to hidden layer:

$$W_h = \sum_{i=1}^d v_{ih} x_i + \theta_h \quad (19)$$

From jth hidden layer to jth hidden layer:

$$W_j = \sum_{h=1}^3 w_{hj} b_h + \theta_j \quad (20)$$

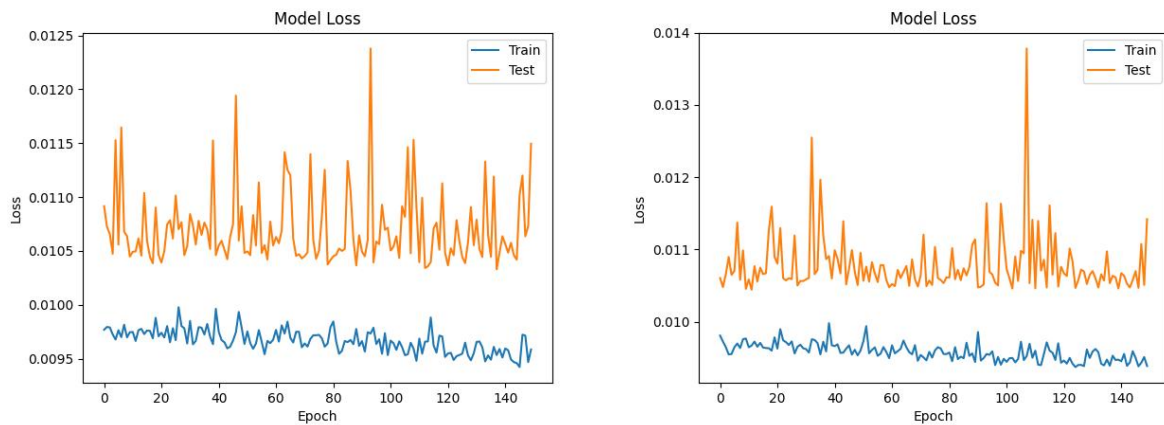
Next, we perform backpropagation, still using "mean squared error" as the loss function. The weights are optimized through the use of Adam, and the final weights are obtained.

6.2.3 Model Comparison

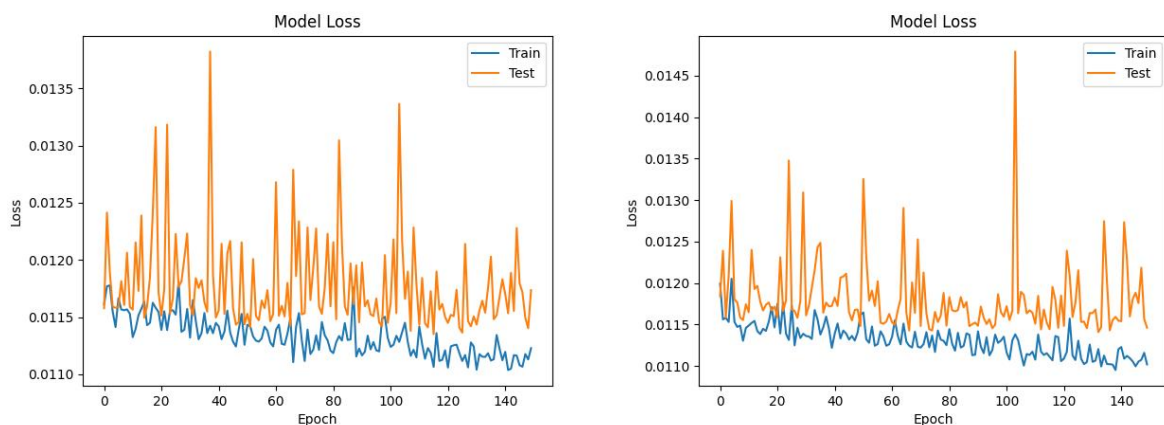
The obtained new variables are brought into the new model and compared with the neural network without introducing dummy variables for regions. The resulting images are shown below.:

Table 7 Comparison of accuracy before and after the introduction of dummy variables

Monohulled Sailboats :



Catamarans:



It can be concluded that the number of outliers is reduced, the final converged loss function is smaller, and the model convergence speed is significantly faster by introducing the region dummy variables. Therefore, the model accuracy is higher after adding the region dummy variables.

Next, the model is trained and the impact of different factors on pricing in the Caribbean, USA, and Europe regions is obtained. The results show in Figure 14.

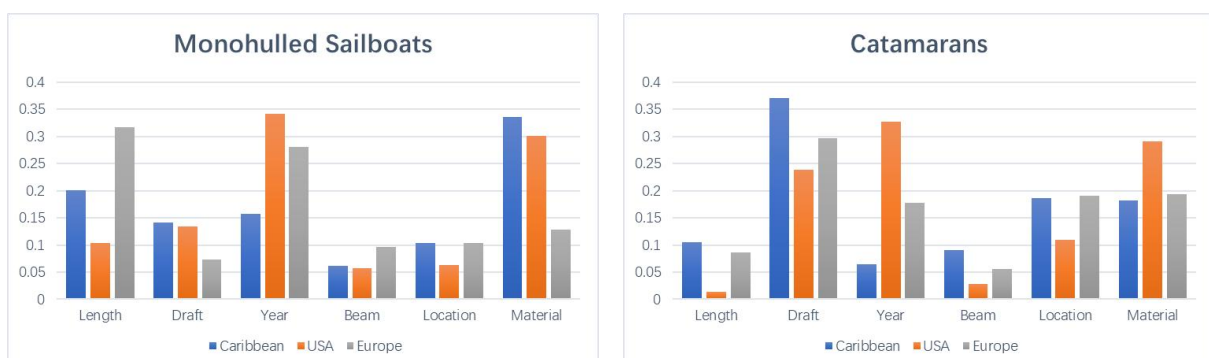


Figure 15 impact of important factors on pricing varies across different regions

From the above chart of Monohulled Sailboats versus Catamarans, it is easy to see that the different types of sailboats are not necessarily influenced by the same region, and we can use the above chart to get an idea of the extent to which the main factors influence the pricing of sailboats in Caribbean, USA and Europe.

6.3 Box plot analysis of price differences:

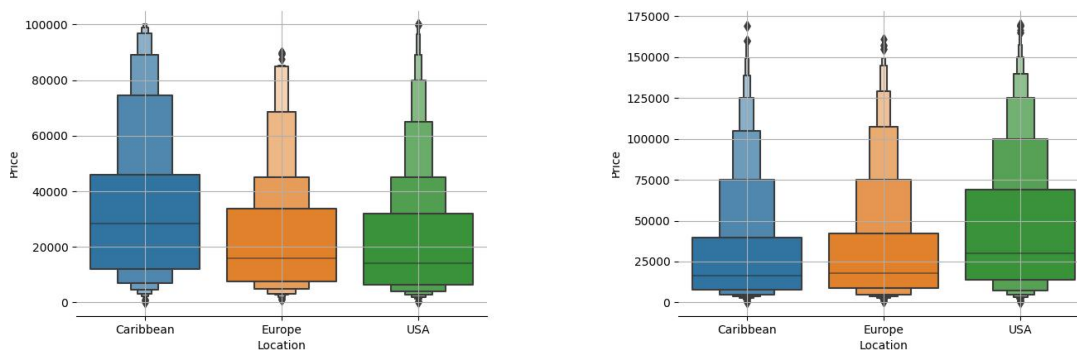


Figure 16 Boxplots show differences in prices across different regions

The results show that Caribbean is the cheapest among Catamarans and the most expensive among Monohulled Sailboats, indicating that people in the Caribbean region are more interested in speed and manoeuvrability, which is closely related to the The Caribbean's proximity to the sea is closely related and supports our conclusions

USA Monohulled Sailboats are the cheapest and Catamarans are the most expensive, indicating that people in the USA focus more on the stability of their boats, which allows them to sail faster downwind and be used for transporting lighter cargo at sea, which also indicates that people in the USA value economic development and justifies our conclusions.

7 Pricing models under the Hong Kong market

7.1 Data collection and processing

First select the information-rich subset of the data given in the question, corresponding to (<http://www.luxboating.com/pre-owned-boat.html>) find the price of the used sailboat corresponding to Year,Length,Beam,Draft,Material, and make a substitution for Location,Price Replace.

Immediately afterwards, the same data pre-processing (invalid data elimination, outlier handling, min-max normalization, etc.) is performed as in Section 4 to make the subsequent model more accurate.

7.2 Establishing a pricing model for Hong Kong

Substituting the processed data into the pricing model in the first question, we can derive the weights of the influence of each factor on the single and double sailing boats in Hong Kong, as shown in the figure below:

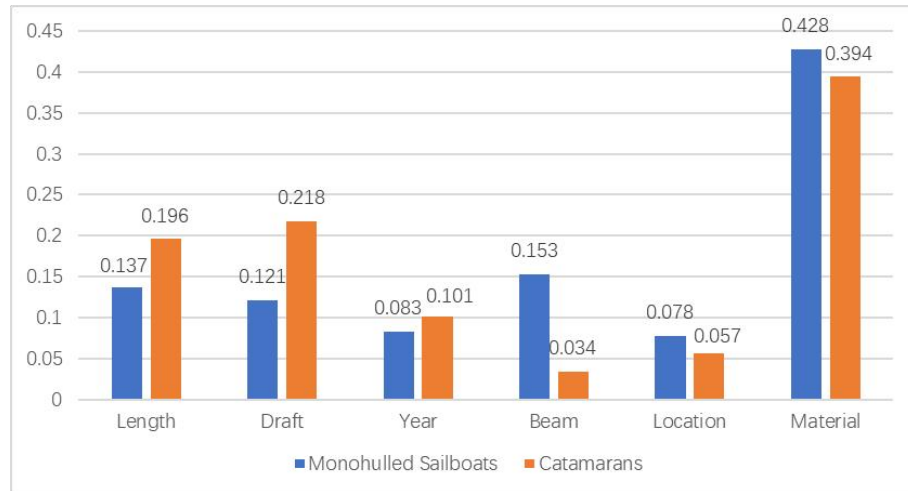


Figure 17 Important factors affect the pricing of sailing boats in Hong Kong

As can be seen from the Figure 18, the Material factor has a strong influence on the pricing of both Monohulled Sailboats and Catamarans in Hong Kong, suggesting that, Hong Kong people place more importance on the material of their sailing boats, which, through analysis, may be due to the fact that there is more sailing competition in Hong Kong and material has a strong influence on boat speed.

Next, we consider the error between the predicted and true values of the model. We first substitute the model, train it and get the predicted values, and get the error of the model by making with the true values. The formula is as follows:

$$E = Y_T - Y_p \quad (21)$$

The error are as follows(Figure 19):

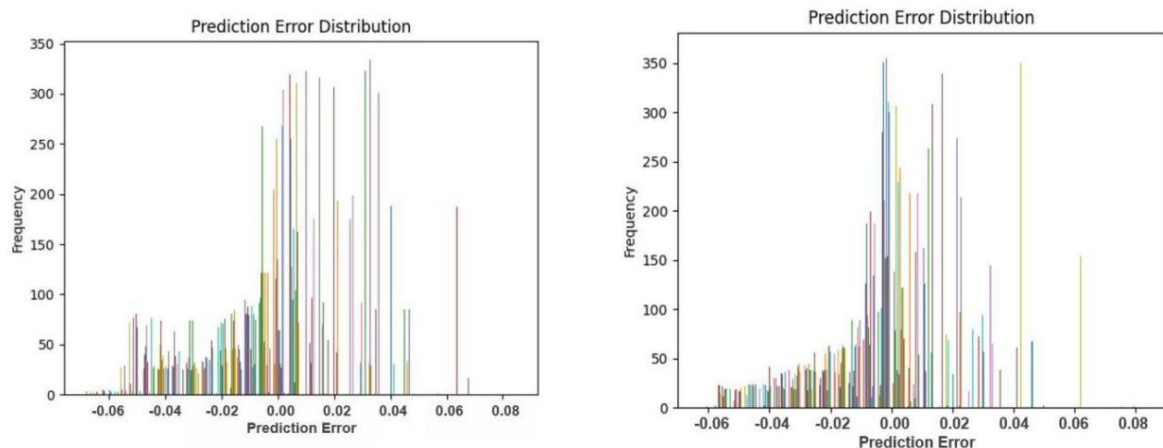


Figure 19 Analysis of errors

It can be clearly seen that the data is between -0.05 and 0.05, indicating that the model predicts Hong Kong with a high degree of accuracy.

8 Interesting and informative inferences or conclusions

- In analyzing the regions of Hong Kong, we found that the pricing of both monohull and catamaran boats mainly depends on the materials used. Based on the characteristics of the region, we analyzed that this is likely due to the prevalence of sailboat racing in Hong Kong, where the material can greatly affect the speed of the sailboat.
- In analyzing the box plots of data from the Caribbean, USA, and Europe, we found that the level of economic development in the region is positively correlated with the prices of catamarans and negatively correlated with the average prices of monohull boats. This may be due to the fact that in regions with a higher level of economic development, sailboats are mainly used for trading purposes.
- In analyzing the pricing of sailboats in the Caribbean, USA, and Europe regions, we found that the same factors affecting pricing show different effects. We found that the main factor affecting the pricing of different types of sailboats is Year. This may be due to the fact that these three regions have a significant aging population, making Year an important pricing factor.

9 Sensitivity Analysis

For our sailboat pricing model, although we have determined the significant parameters through factor analysis, we still need to identify which of the six factors is most important to potential buyers. Additionally, we introduced dummy variables in the second question to improve the stability of the model.

Therefore, we conducted sensitivity analysis on both models, and the results are shown in the following figure:

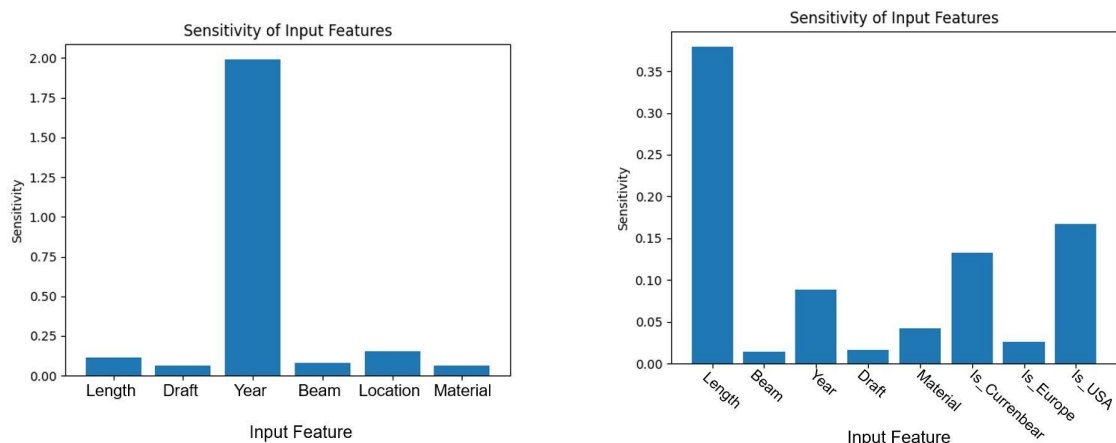


Figure 21 9 Sensitivity Analysis chart

In the first model, the sailboat's price is highly sensitive to the year of manufacture. This suggests that the year of manufacture is a crucial factor for potential buyers, as older boats are more likely to have maintenance issues and may exceed their budget for upkeep. This finding indicates that the model aligns well with real-world scenarios.

For the second model, the introduction of dummy variables has reduced the sensitivity to the year of manufacture, as well as the sensitivity of all parameters to below 0.4. This indicates that the model has achieved its intended goal of stability. Additionally, the sensitivity of Europe is lower than the other two regions, which is consistent with the box plot analysis mentioned earlier, where Europe appears to be in the middle position in all scenarios.

10 Model Evaluation and Further

10.1 Strengths

- (1) The backpropagation algorithm is an effective optimization technique that can automatically adjust the weights and biases of a BP neural network to fit the sample data. This enables the BP neural network to be trained using the backpropagation algorithm, allowing for accurate regression analysis.
- (2) The sailing parameters and pricing relationships are often unknown, and the presence of both linear and non-linear relationships may not be evident. In such cases, a regression model using a BP neural network is a suitable approach for a wide range of regression problems, including one-dimensional or multiple regression, linear or non-linear regression. This method is capable of achieving a high level of accuracy and obtaining a good fit even when dealing with complex relationships.
- (3) Factor analysis is a powerful tool for identifying the interrelationships between different characteristic factors of a sailboat. By analyzing how certain parameters influence others, it can provide insights into the complex relationships between sailboat parameters. This understanding can help in selecting a better combination of sailing yacht pricing parameters, and ultimately, improve the overall accuracy of the pricing model.

10.2 Weaknesses

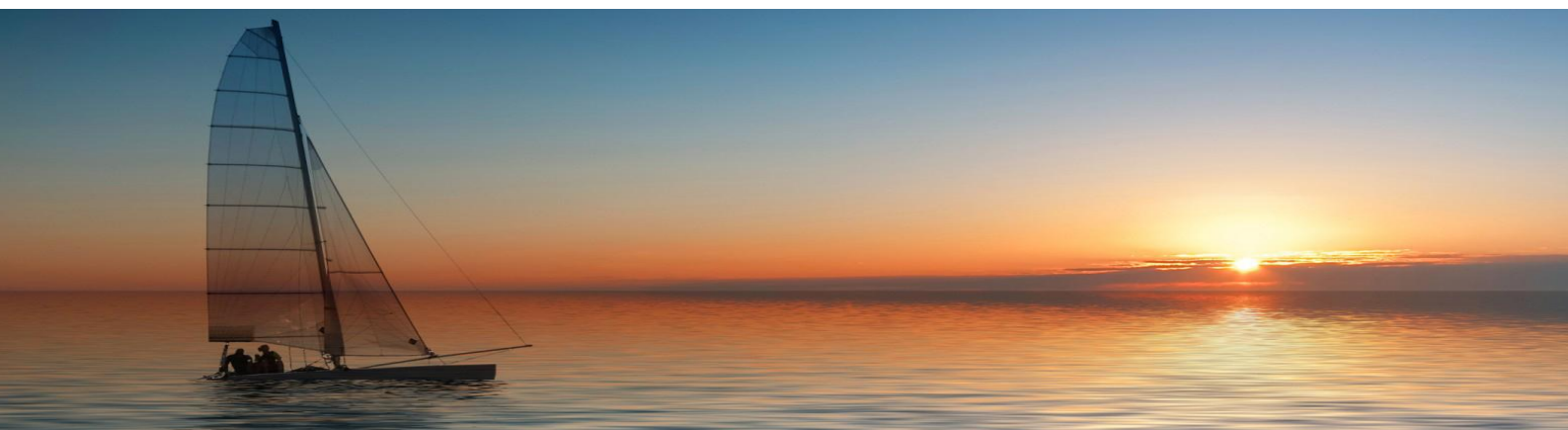
- (1) The factors obtained from factor analysis are a result of linear combinations of the original variables, and they do not hold a specific practical meaning on their own. Additionally, there may be multiple possible outcomes.
- (2) Adequate training data is required for bp neural network regression to prevent overfitting and ensure accurate predictions.

10.3 Further Discussion

When using the Sigmoid activation function to obtain the bp neural network regression model, there may be issues of gradient saturation and gradient disappearance. To address these issues, improved activation functions such as ReLU and LeakyReLU can be utilized to enhance the approximation ability and training speed of the network.

References

- [1] Source:<http://yachter123.com/sy>
- [2] Zhang Shufan,Zhang String.Application of BP neural network in second-hand ship valuation[J]. China Asset Appraisal,2015(11):21-25.
- [3] Xiao Qijun, Zhang Yanmeng. Based on BP neural network of secondhand ship price assessment study [J]. Journal of Marine engineering, 2013, 35 (02) : 100-103. The DOI: 10.13788 / j.carol carroll nki CBGC. 2013.02.008.
- [4] Cruising World. (2007, February). Pricing Used Sailboats. Retrieved from <https://www.cruisingworld.com/sailboats/pricing-used-sailboats/>.
- [5] BoatUSMagazine. (2015, May). Factors Affecting the Price of Used Boats.Retrieved from.
<https://www.boatus.com/magazine/2015/may/factors-affecting-the-price-of-used-boats>.
- [6] Power & Motoryacht. (2019, August). Boat Buying Guide: How to Price a Boat. Retrieved from<https://www.powerandmotoryacht.com/maintenance/boat-buying-guide-how-to-price-a-boat>.
- [7] Boating. (2018, November). Used Boat Values: What They’re Really Worth. Retrieved from
<https://www.boatingmag.com/boats/used-boat-values-what-theyre-really-worth/>.
- [8] Layard, R., Mayraz, G., & Nickell, S. (2008). The marginal utility of income. Journal of Public Economics, 92(8-9), 1846-1857. doi: 10.1016/j.jpubeco.2008.04.004
- [9] Bi Xiaowei, Liu Manman, Fang Dongsheng, Li Yang. Target threat ranking method based on factor analysis [J]. Electronic Information Countermeasures Technology, 2023,38 (02): 1-7
- [10] Zheng Yue, Chen Liwei, Sun Delong, Zhang Anxiao. Study on the influencing factors of China's architectural heritage protection and management based on BP-MIV [J]. New Architecture, 2022 (06): 82-86
- [11] Yang Bo. Research on Used Car Evaluation and Pricing Model in the "Internet plus" Era Based on Intelligent Algorithms [J]. Jiangsu Business Review, 2017 (05): 21-24. DOI: 10.13395/j.cnki.issn.1009-0061.2017.05.005



A report to a Hong Kong broker

——Used Sailboat Pricing Model

Hong Kong has over 200 islands and numerous good harbors. Every year, many international sailing competitions are held, and the sailing market has a good prospect in Hong Kong. The price of sailboats varies depending on various factors such as the type of boat. It is worth considering how to accurately determine the pricing of sailboats to maximize profits.

We provide accurate pricing for brokers in the Hong Kong sailing market by developing a BP neural network based regression pricing model for used sailing boats. we applied neural network regression to assess the weight of each factor affecting pricing between single-hull and double-hull vessels.



Visualising the Hong kong weight data, it can be seen that the biggest factor influencing the pricing of used boats is hull material, which is likely to be related to the large number of international regattas that take place in Hong Kong each year as material is the main factor influencing the speed of the boat. Secondly, hull length and width also have a significant effect on pricing, most likely due to the fact that there is a limit to the length and width of a hull for sailing competitions.

According to the findings of this study, it is advisable for sailing brokers operating in Hong Kong (SAR) to closely monitor developments in the sailing market, particularly changes in hull material, length, and width, when determining the pricing strategy for their sailing boats. The study highlights that compared to other factors, the length, draft, and material ratio holds greater significance, with a high rating of 0.7-0.8. As a result, it is suggested that brokers consider investing in race-worthy sailing boats with these specific attributes, but at a lower cost, as this could result in increased profitability.

Appendices

Appendix 1

Introduce: Pricing model code

```
import numpy as np
import pandas as pd
from keras.models import Sequential
from keras.layers import Dense
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
import matplotlib.pyplot as plt

# Read data
data = pd.read_excel('boats_data 1_1.xlsx')
X = data.iloc[:, 1:7].values
y = data.iloc[:, 0].values
# Data standardisation
scaler = StandardScaler()
X = scaler.fit_transform(X)
# Splitting the dataset into a training set and a test set
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=0)
# print(X_train)
# Building the model
model = Sequential()
model.add(Dense(units=50, activation='sigmoid', input_dim=6))
model.add(Dense(units=50, activation='sigmoid'))
model.add(Dense(units=50, activation='sigmoid'))
model.add(Dense(units=1))
# model.add(Dense(units=64, activation='relu', input_dim=8))
# model.add(Dense(units=64, activation='relu'))
# model.add(Dense(units=1))
# Compilation model
model.compile(loss='mean_squared_error', optimizer='adam')

#categorical_crossentropy      binary_crossentropy      mean_absolute_error
mean_squared_error

# Training models
model.fit(X_train, y_train, epochs=150, batch_size=32, validation_data=(X_test, y_test))
```

```
# Using the test set for prediction
y_pred = model.predict(X_test)

# print(y_pred)

# Loss function
history = model.fit(X_train, y_train, epochs=150, batch_size=32,
validation_data=(X_test, y_test))
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
plt.title('Model Loss')
plt.ylabel('Loss')
plt.xlabel('Epoch')
plt.legend(['Train', 'Test'], loc='upper right')
plt.show()
# Output final weights

weights = model.get_weights()
#Get the total weight value of the respective variable
variable1_total_weight = sum(weights[0][0])
variable2_total_weight = sum(weights[0][1])
variable3_total_weight = sum(weights[0][2])
variable4_total_weight = sum(weights[0][3])
variable5_total_weight = sum(weights[0][4])
variable6_total_weight = sum(weights[0][5])
# variable7_total_weight = sum(weights[0][6])
# variable8_total_weight = sum(weights[0][7])
print(variable1_total_weight)
print(variable2_total_weight)
print(variable3_total_weight)
print(variable4_total_weight)
print(variable5_total_weight)
print(variable6_total_weight)
# print(variable7_total_weight)
# print(variable8_total_weight)
```